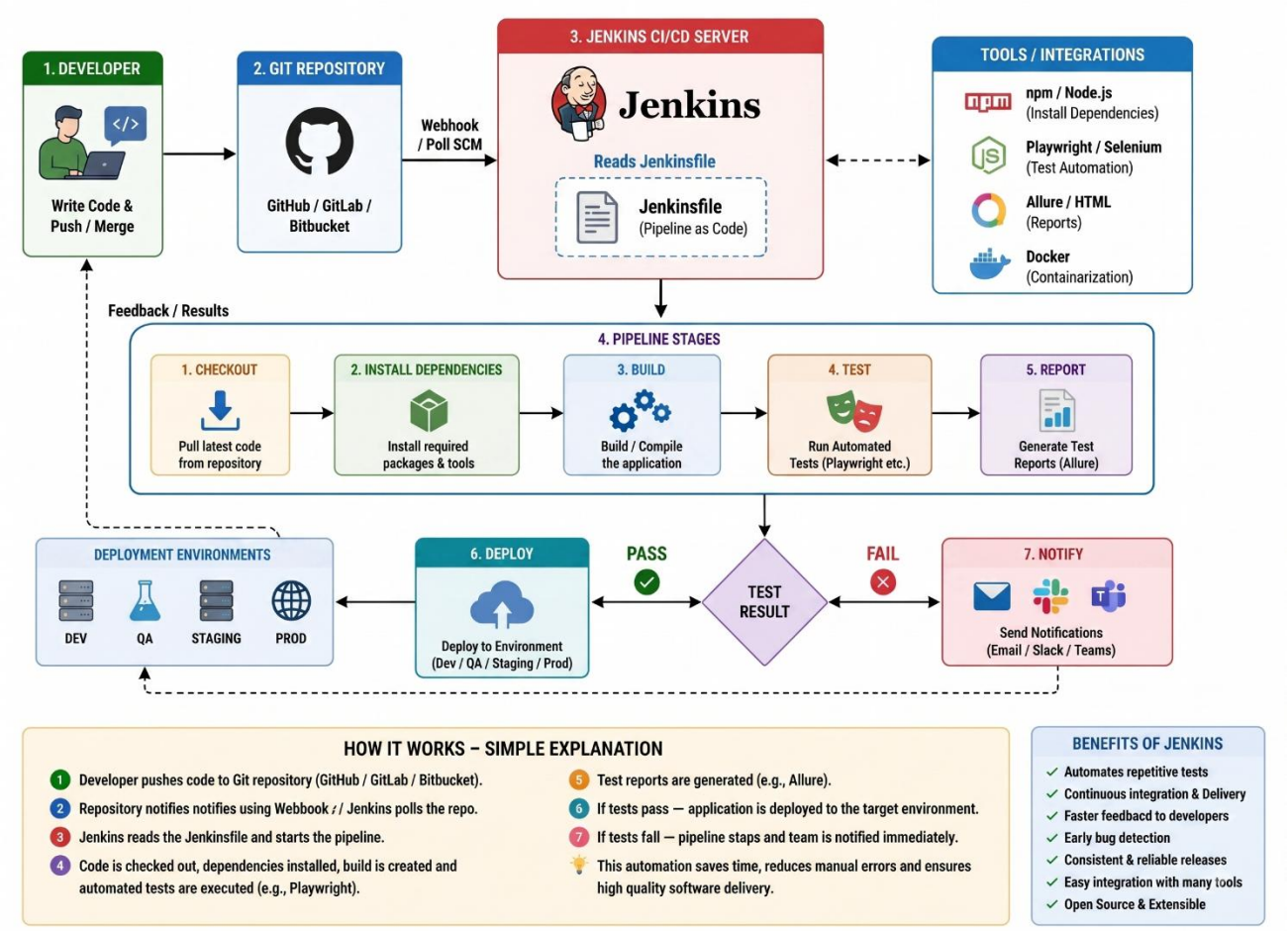
Jenkins CI/CD Pipeline



HOW IT WORKS – SIMPLE EXPLANATION

1. Developer pushes code to Git repository (GitHub / GitLab / Bitbucket).
2. Repository notifies notifies using Webbook / Jenkins polls the repo.
3. Jenkins reads the Jenkinsfile and starts the pipeline.
4. Code is checked out, dependencies installed, build is created and automated tests are executed (e.g., Playwright).
5. Test reports are generated (e.g., Allure).
6. If tests pass — application is deployed to the target environment.
7. If tests fall — pipeline staps and team is notified immediately.
- This automation saves time, reduces manual errors and ensures high quality software delivery.

BENEFITS OF JENKINS

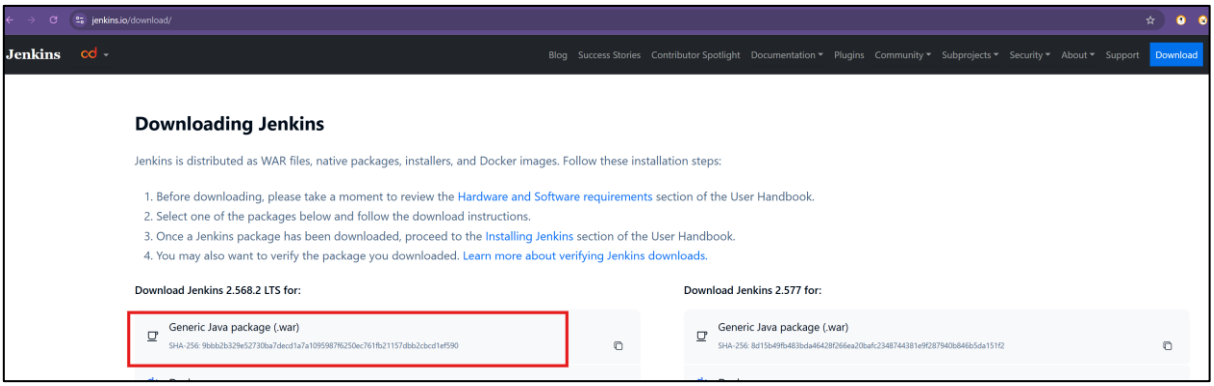
- ✓ Automates repetitive tests
- ✓ Continuous integration & Delivery
- ✓ Faster feedbacd to developers
- ✓ Early bug detection
- ✓ Consistent & reliable releases
- ✓ Easy integration with many tools
- ✓ Open Source & Extensible

Installation

Download .war file

https://www.jenkins.io/download/

**** JDK Should be installed to run Jenkins.**

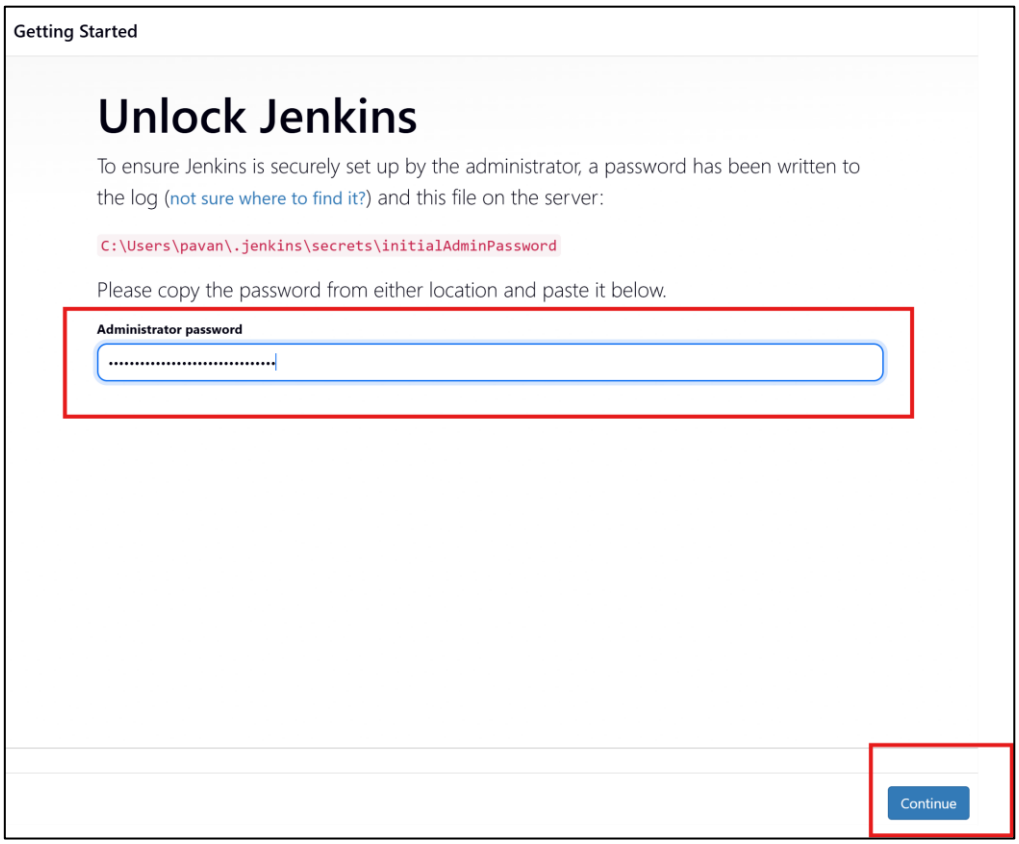


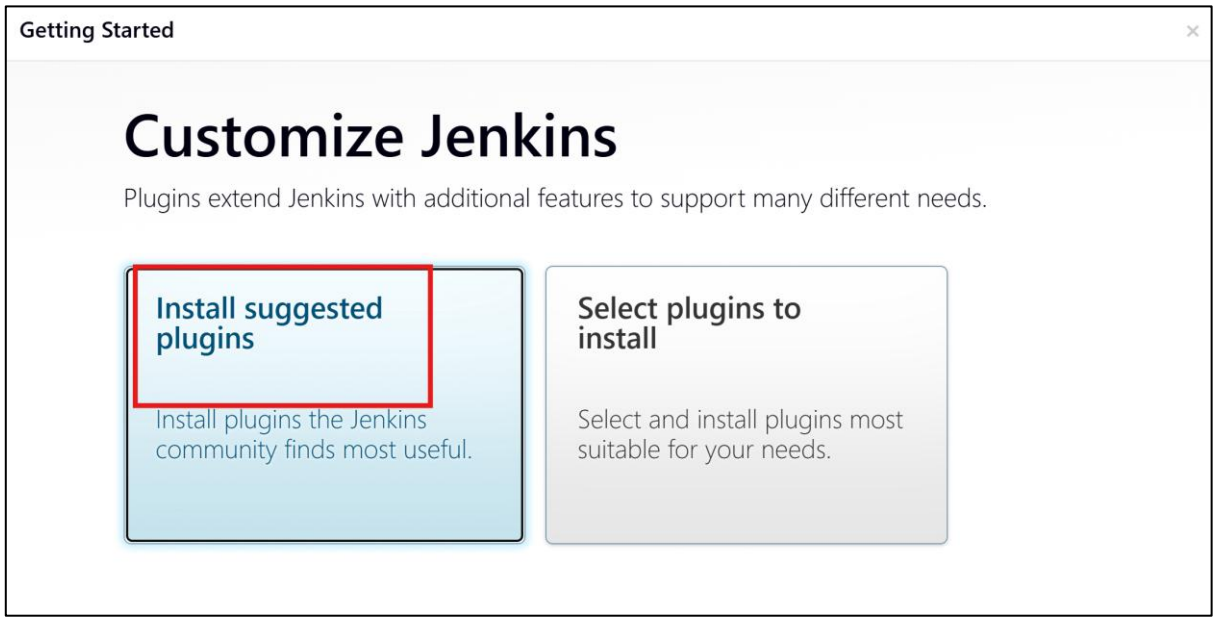


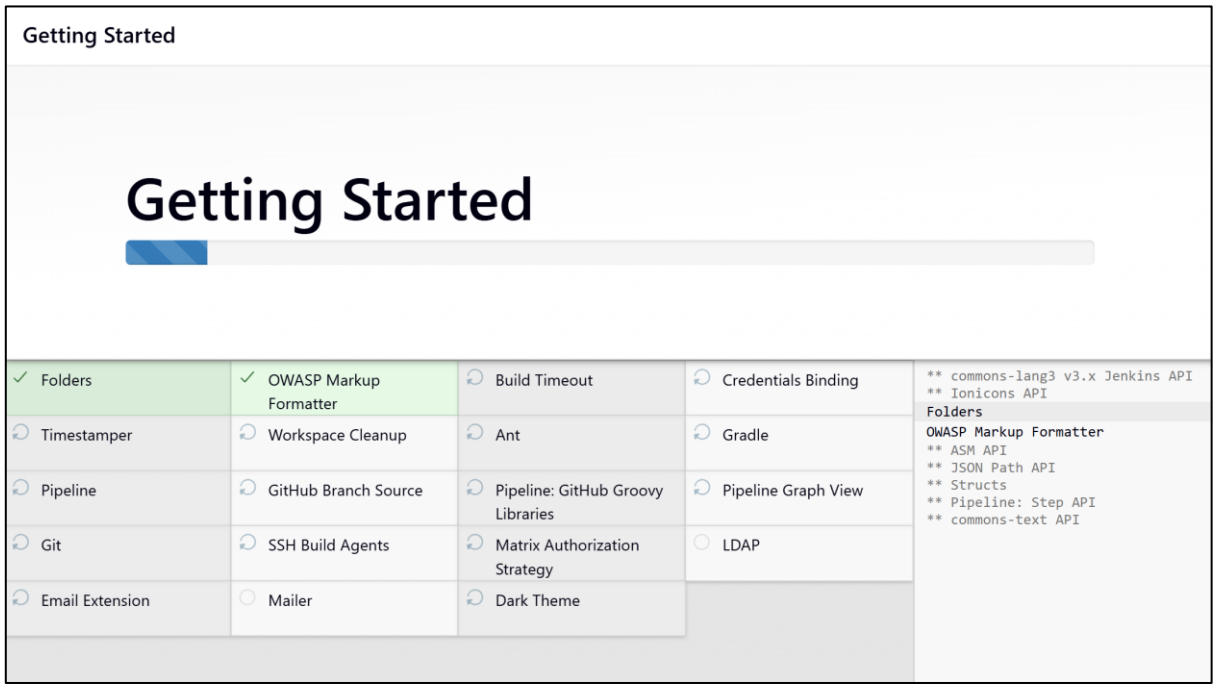
Run Jenkins.war file using Command Line/terminal.

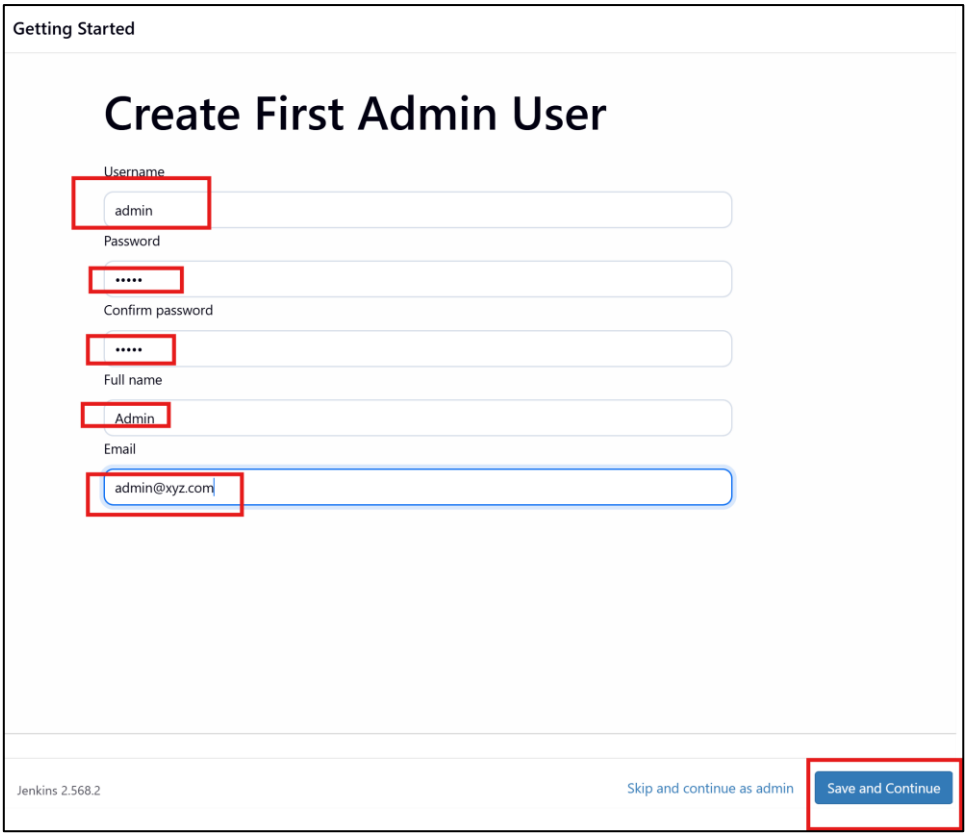
```
C:\Automation>java -jar jenkins.war --enable-future-java
Running with Java 22 from C:\Program Files\Java\jdk-22, which is not fully supported. Continuing because --enable-future-java is set. Supported Java ve
ps://jenkins.io/redirect/java-support/ for more information.
Running from: C:\Automation\jenkins.war
webroot: C:\Users\pavan\.jenkins\war
2026-08-16 07:24:25.973+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-08-16 07:24:26.828+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-16 07:24:26.876+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa4
9; jvm 22.0.2+9-70
2026-08-16 07:24:27.416+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jett
2026-08-16 07:24:27.459+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-08-16 07:24:27.807+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\pavan\.jenkins found at: $user.h
2026-08-16 07:24:27.889+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3f93e4a8{Jenki
ers/pavan/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@12b5454f{STARTED}}
2026-08-16 07:24:27.931+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@64412d34{HTTP/1.1, (http/1.1)}{0.0
2026-08-16 07:24:27.931+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@38c2c309{STARTING}[12.1.8,sto=0] @1054ms
2026-08-16 07:24:27.931+0000 [id=56]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-16 07:24:28.013+0000 [id=52]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.568.2
2026-08-16 07:24:28.090+0000 [id=64]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-16 07:24:28.096+0000 [id=84]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-16 07:24:28.784+0000 [id=92]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-16 07:24:28.784+0000 [id=64]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-16 07:24:28.784+0000 [id=72]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-16 07:24:28.929+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-16 07:24:28.936+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-16 07:24:28.938+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-16 07:24:28.938+0000 [id=87]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-16 07:24:28.955+0000 [id=107]   INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-16 07:24:29.189+0000 [id=66]    INFO    jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> *************************************************************
[CRLF]> *************************************************************
[CRLF]> *************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> f1891fa8348948a79939e7a8c907dc11
[CRLF]>
[CRLF]> This may also be found at: C:\Users\pavan\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> *************************************************************
[CRLF]> *************************************************************
```

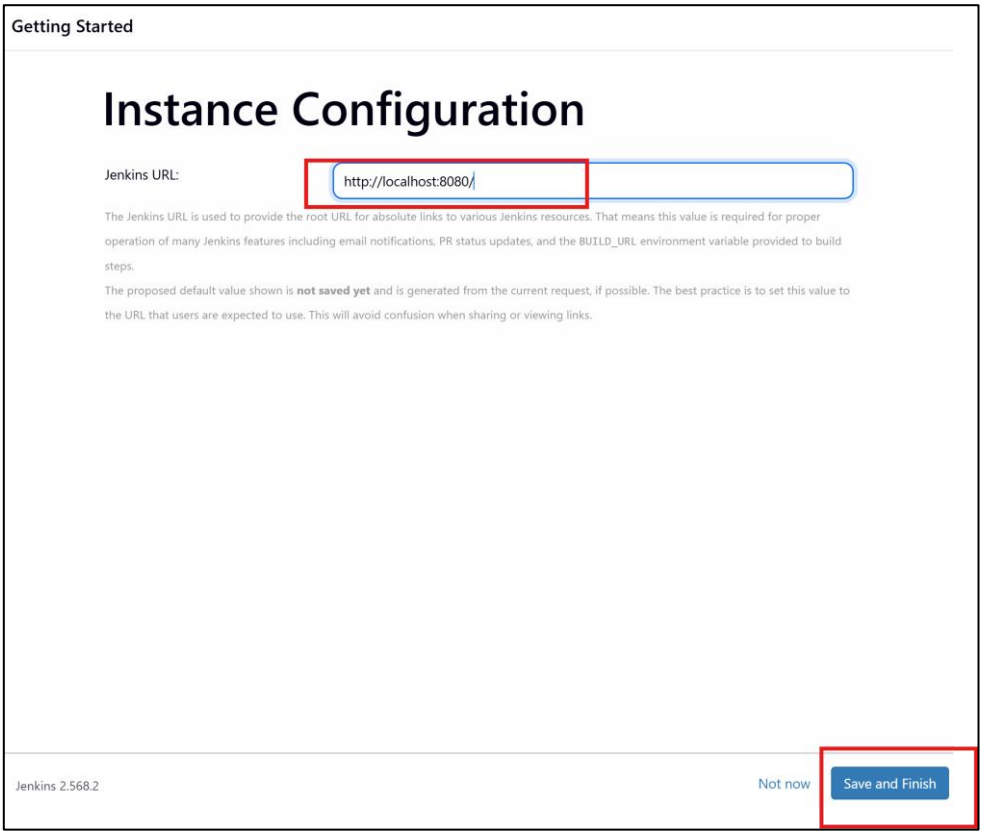
http://localhost:8080/




Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.

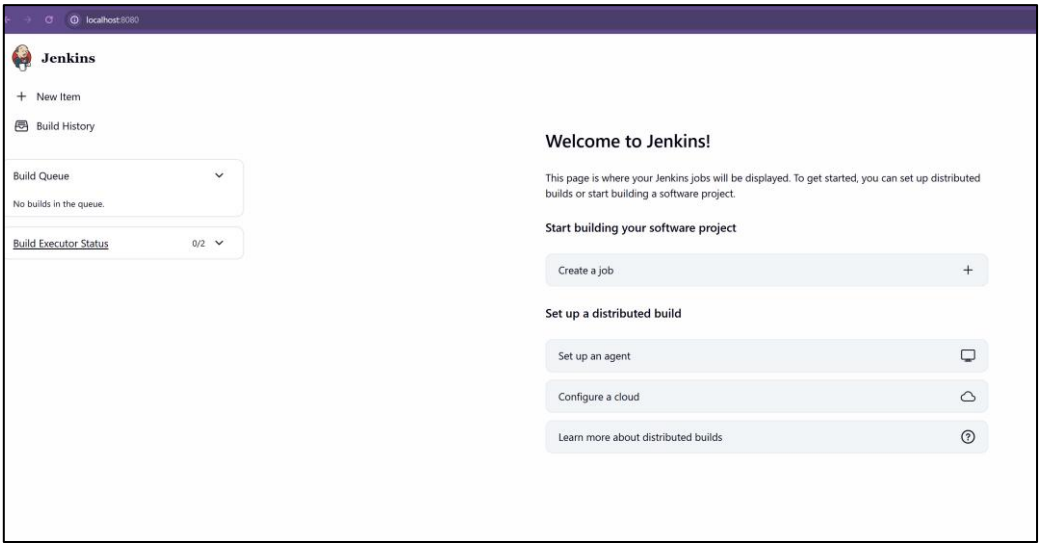

Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** JSON Path API
** Structs
** Pipeline: Step API
** commons-text API


Getting Started
Create First Admin User
Username
admin
Password
Confirm password
Full name
Admin
Email
admin@xyz.com
Jenkins 2.568.2
Skip and continue as admin
Save and Continue

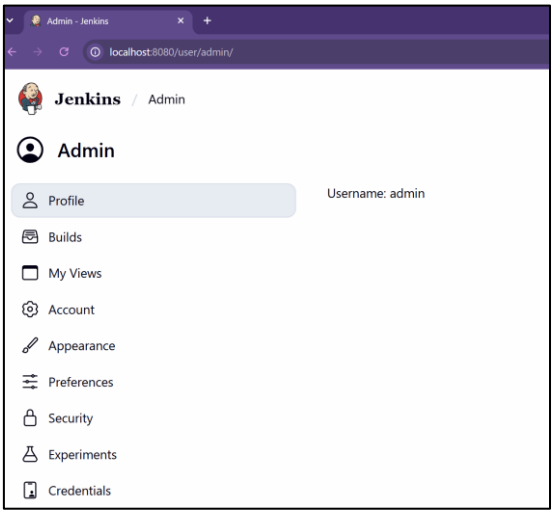

Getting Started
Instance Configuration
Jenkins URL:
http://localhost:8080/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.568.2
Not now
Save and Finish



Jenkins First Page:



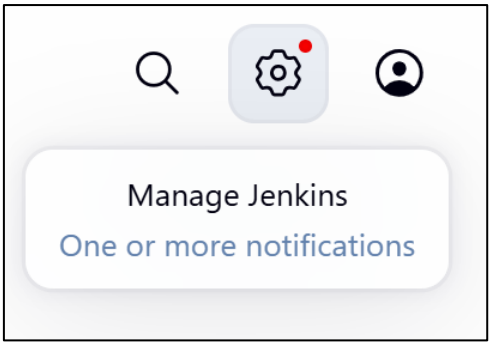
Dashboard Page:



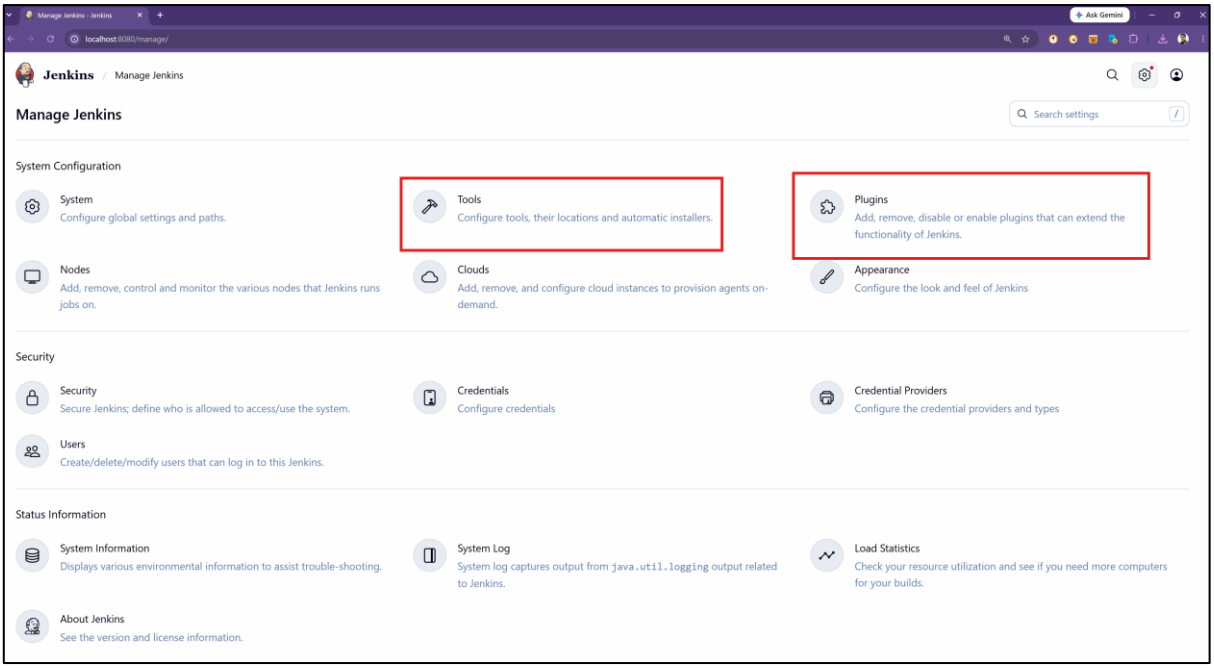
Jenkins One Time Settings

Install required Jenkins Plugins:

Plugin	Used For
Pipeline	Declarative pipeline syntax
Git	checkout scm
Timestamper	options { timestamps() }
PowerShell	powershell steps
JUnit	junit testResults
HTML Publisher	publishHTML step
Allure Jenkins Plugin	allure step
NodeJS Plugin *(recommended)*	Manage Node.js via Jenkins Tools

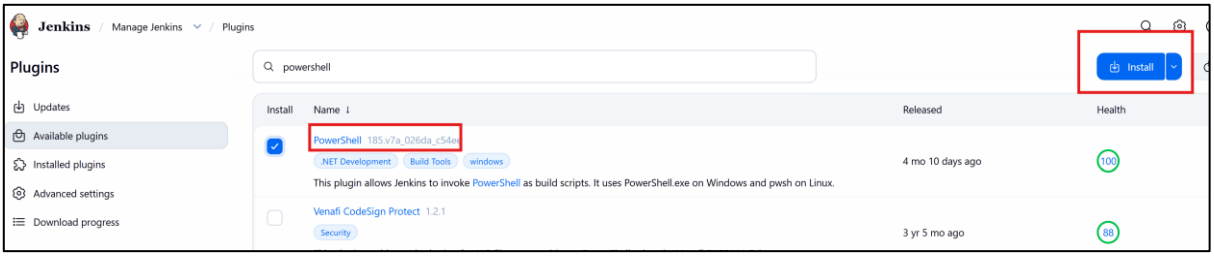


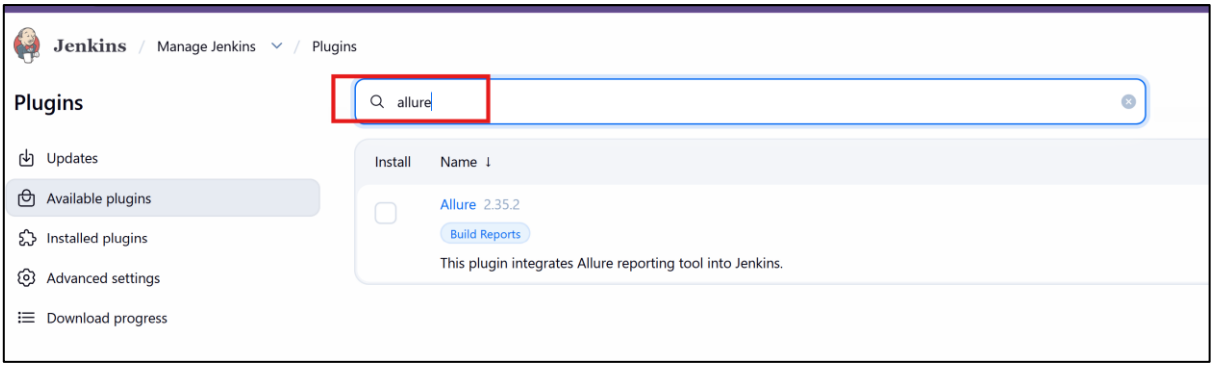
Navigate to Manage Jenkins →Plugins

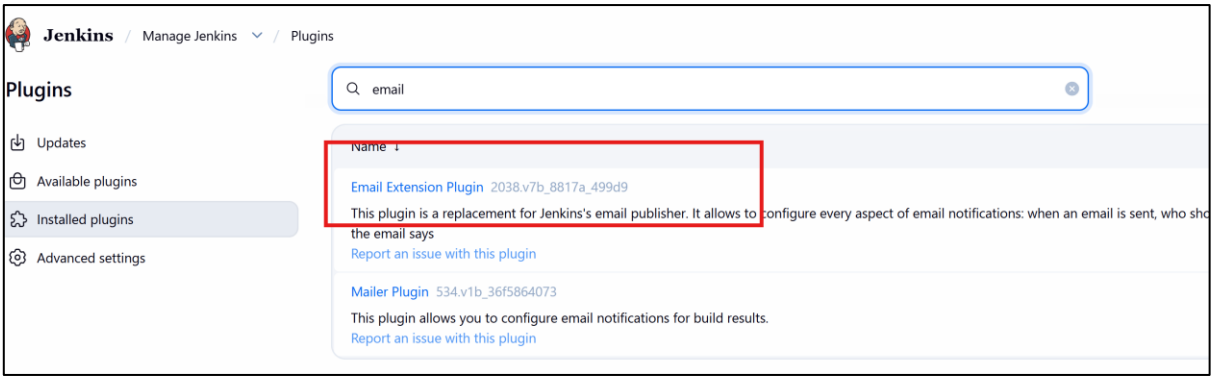


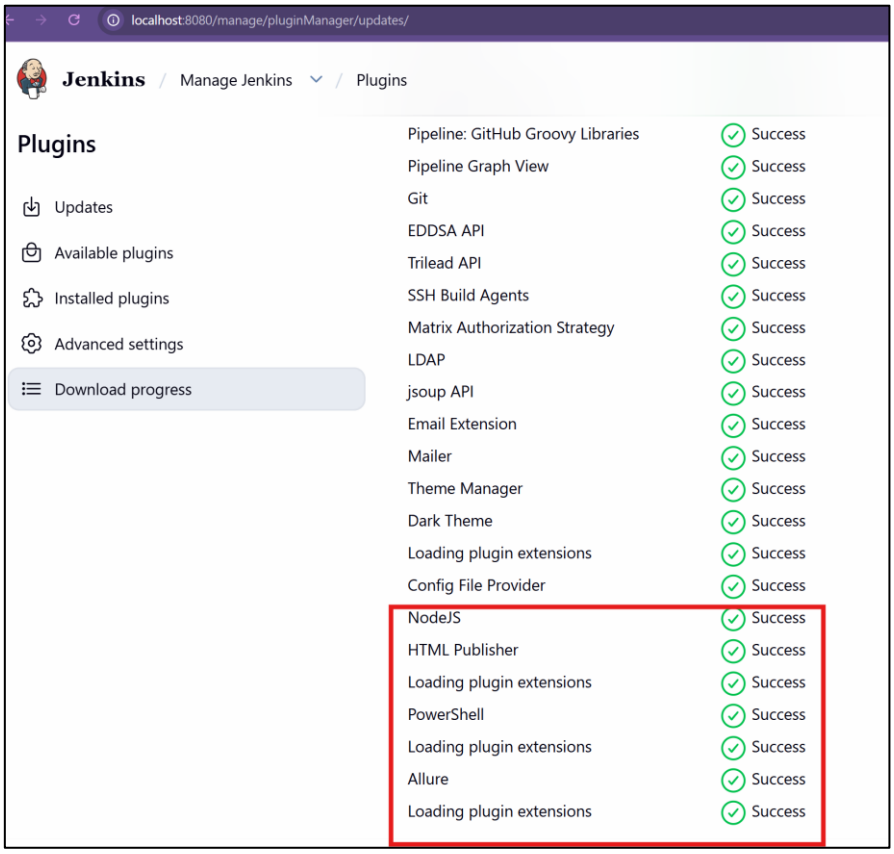
Install required additional plug-ins







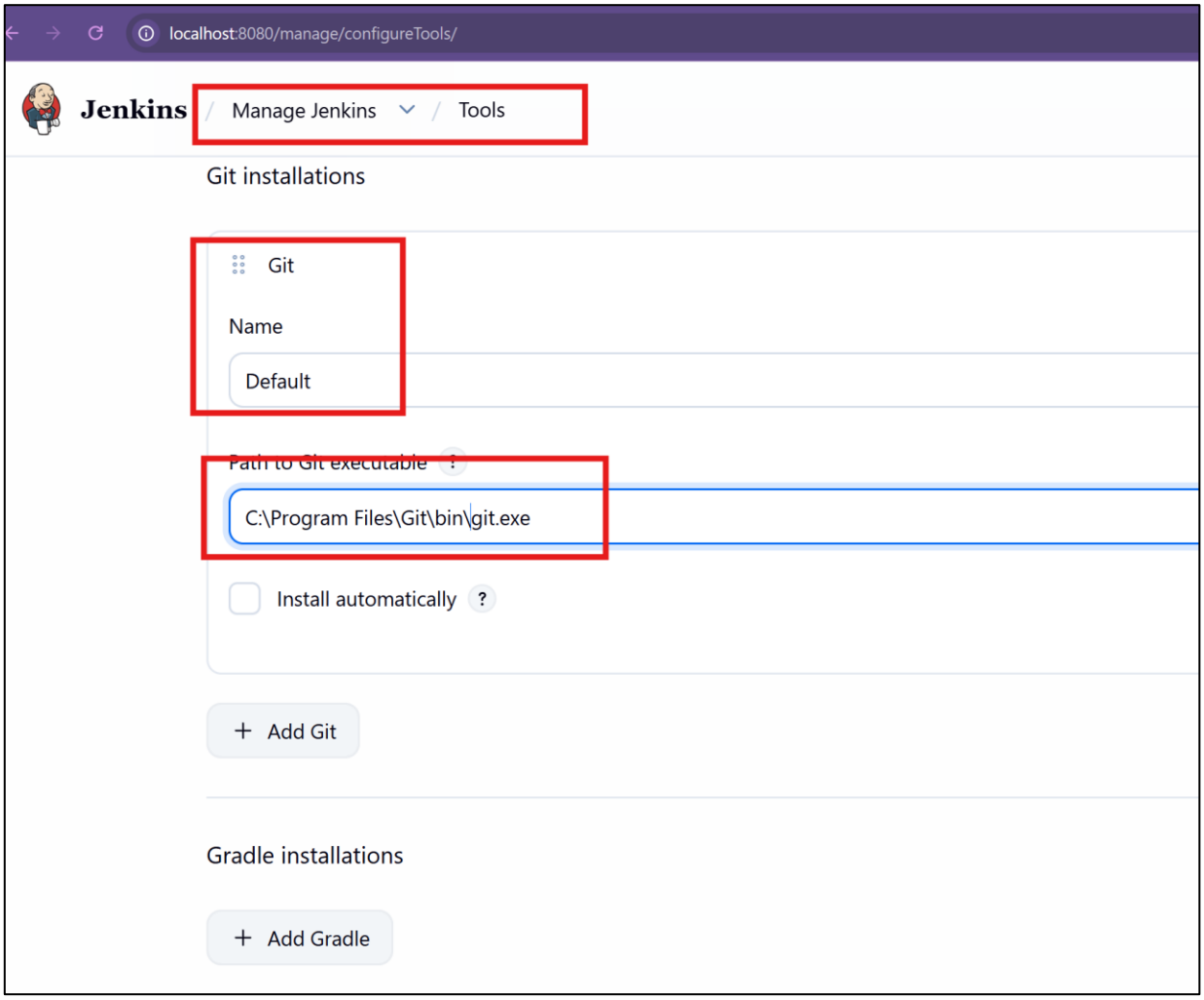

Jenkins / Manage Jenkins / Plugins
Plugins
email
Updates
Available plugins
Installed plugins
Advanced settings
Name
Email Extension Plugin 2038.v7b_8817a_499d9
This plugin is a replacement for Jenkins's email publisher. It allows to configure every aspect of email notifications: when an email is sent, who sho
the email says
Report an issue with this plugin
Mailer Plugin 534.v1b_36f5864073
This plugin allows you to configure email notifications for build results.
Report an issue with this plugin


localhost:8080/manage/pluginManager/updates/
Jenkins / Manage Jenkins / Plugins
Plugins
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
Pipeline: GitHub Groovy Libraries Success
Pipeline Graph View Success
Git Success
EDDSA API Success
Trilead API Success
SSH Build Agents Success
Matrix Authorization Strategy Success
LDAP Success
jsoup API Success
Email Extension Success
Mailer Success
Theme Manager Success
Dark Theme Success
Loading plugin extensions Success
Config File Provider Success
NodeJS Success
HTML Publisher Success
Loading plugin extensions Success
PowerShell Success
Loading plugin extensions Success
Allure Success
Loading plugin extensions Success

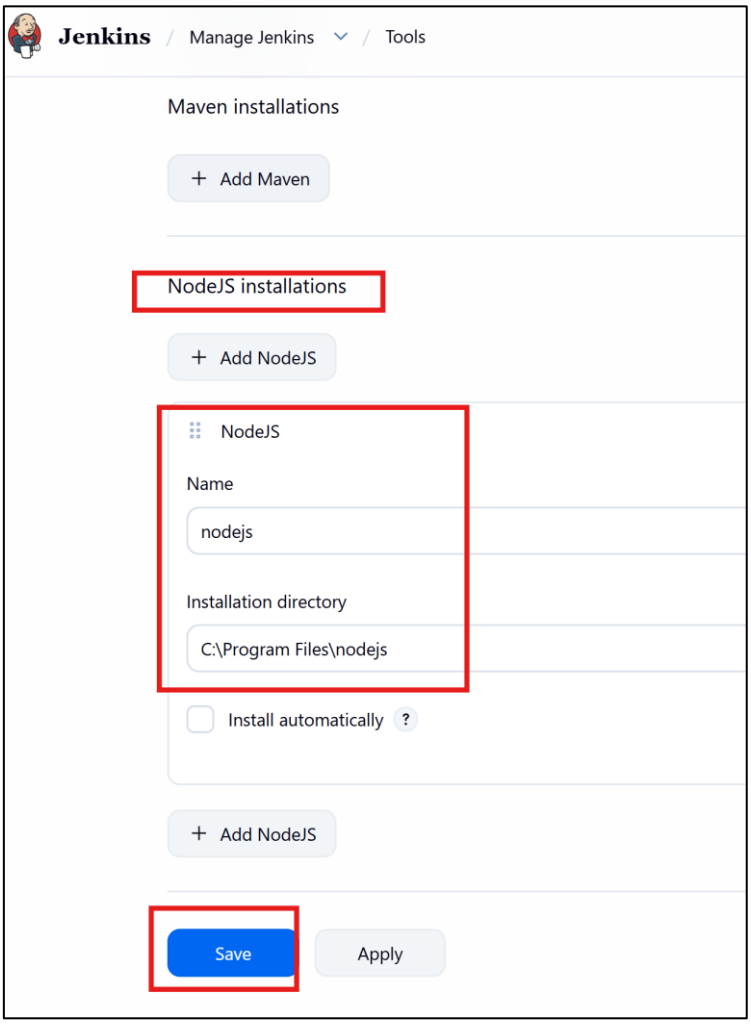
Tools Configuration:

Manage Jenkins → Tools

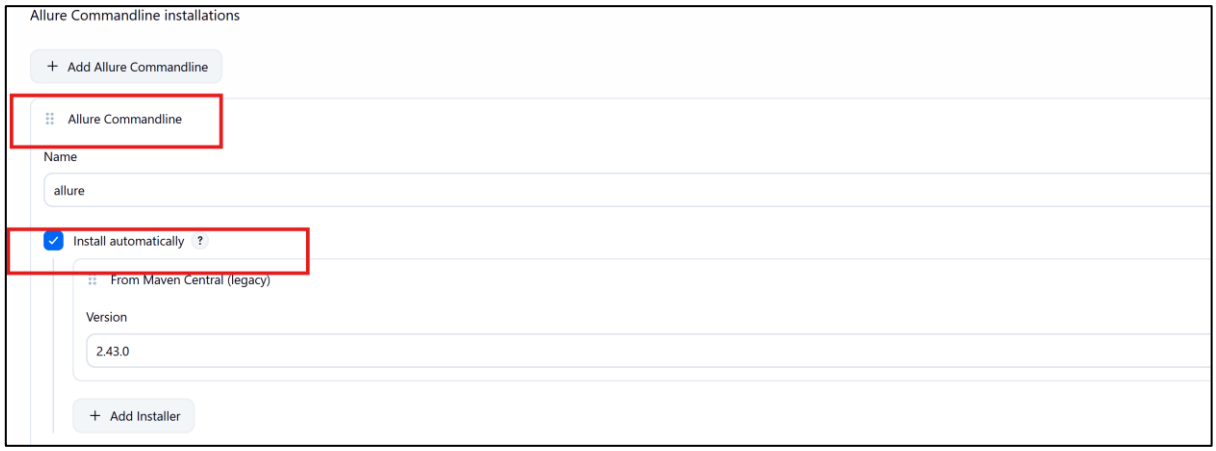
Git:



NodeJS:

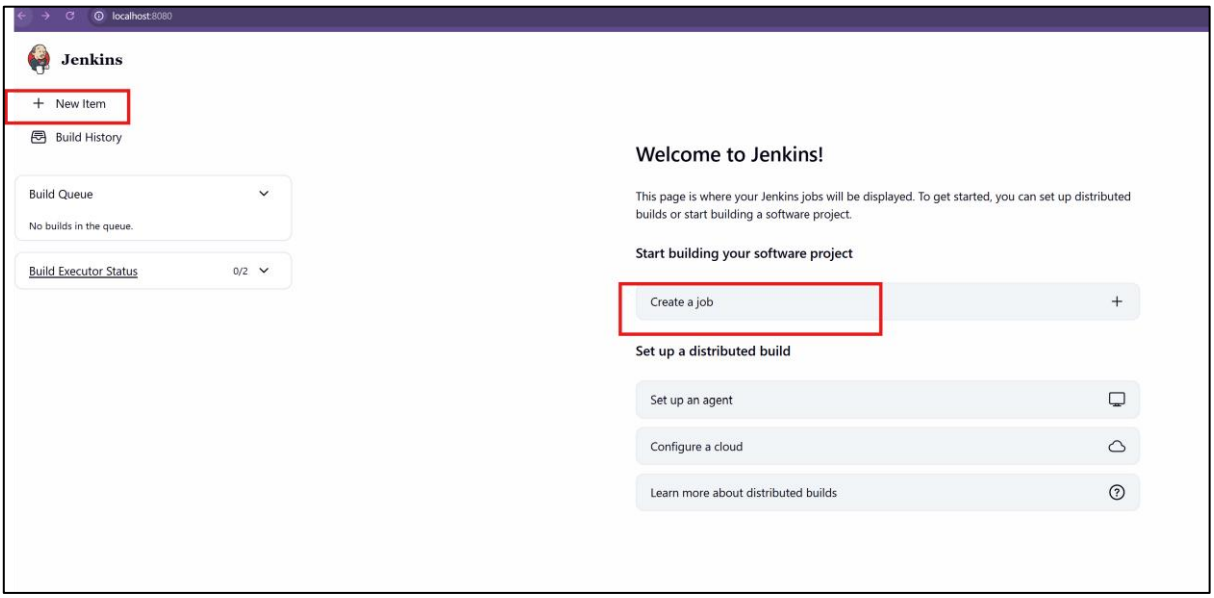


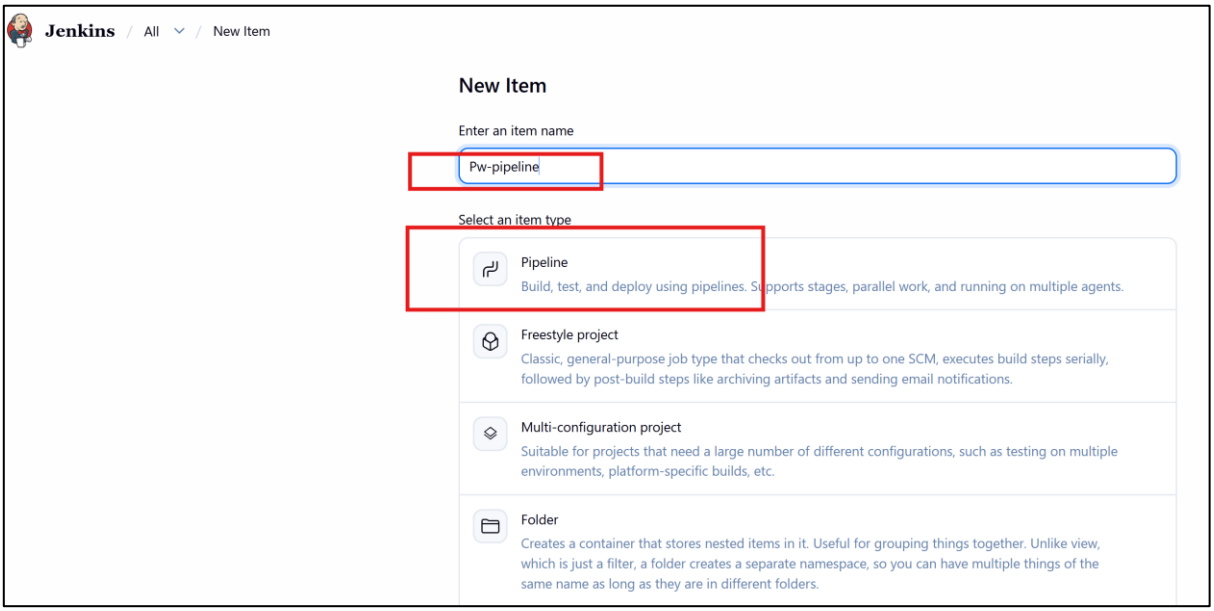
Allure:

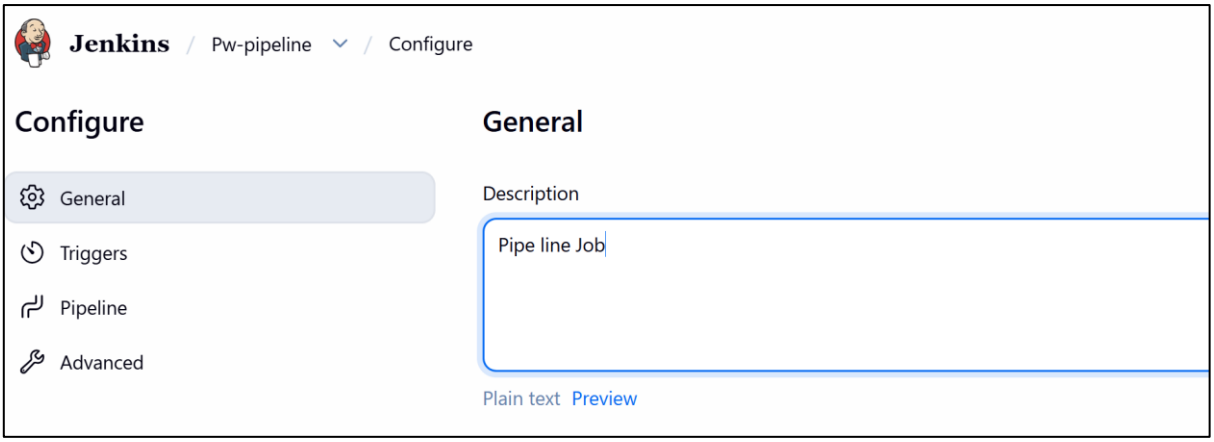


Creating Pipeline Job and Scheduling the Job

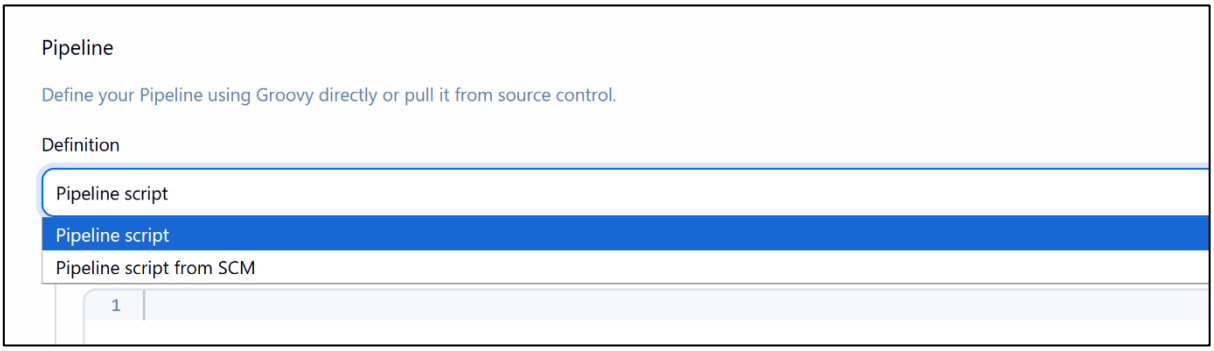
Click on New Item:



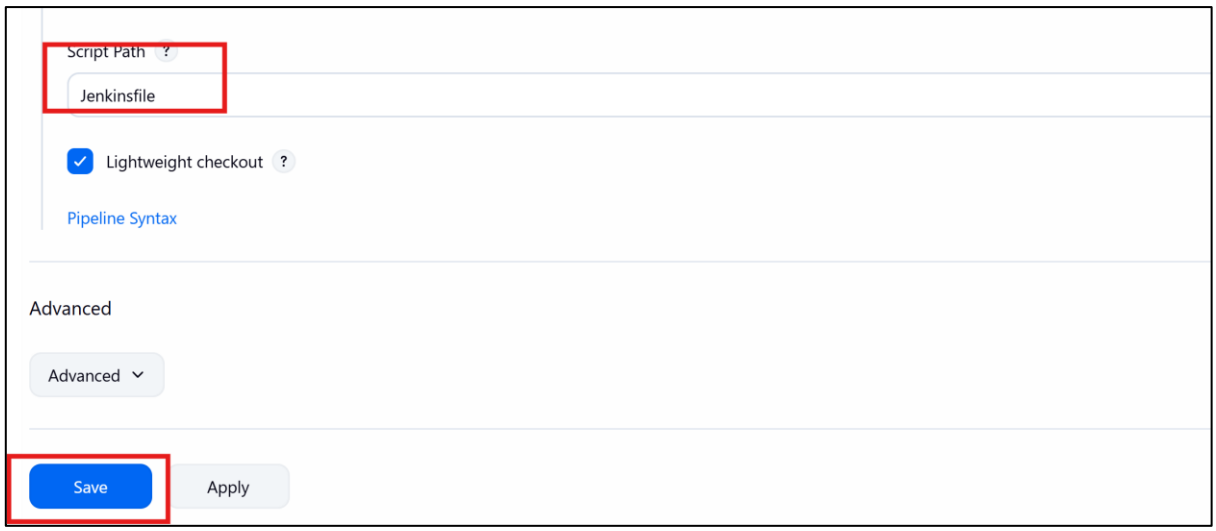




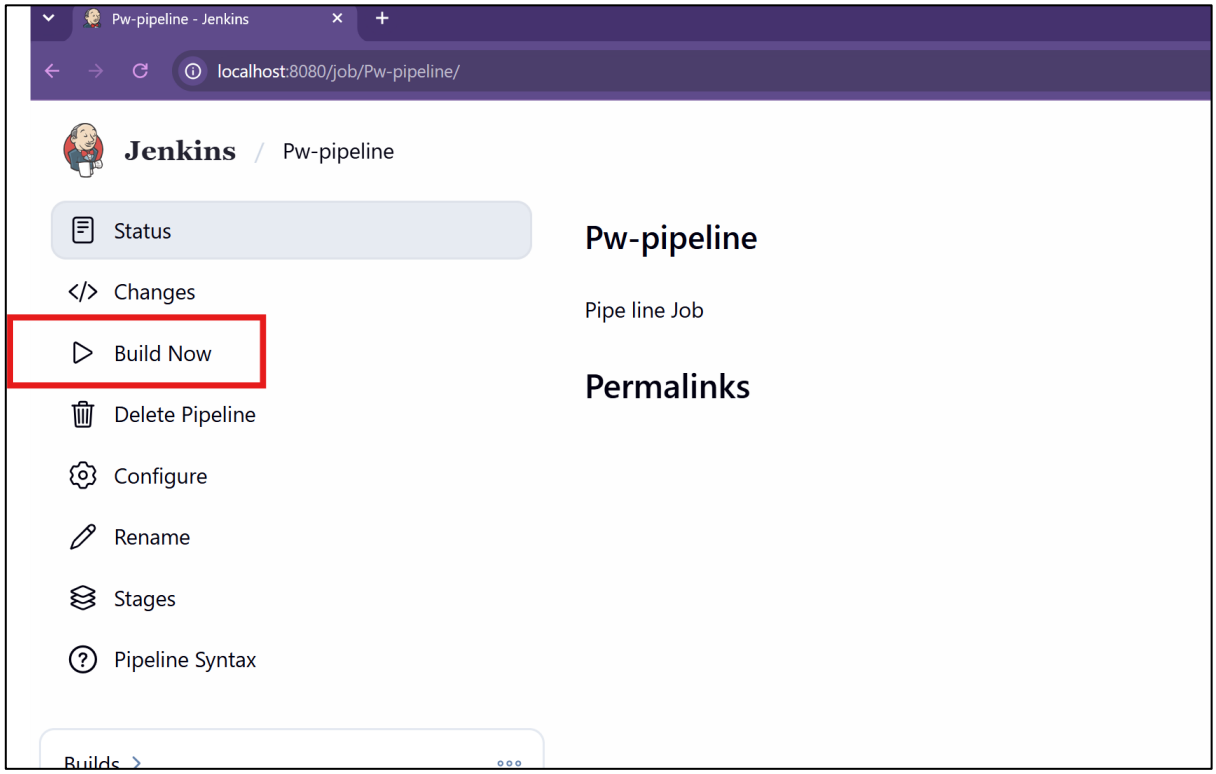
Pipeline creation:

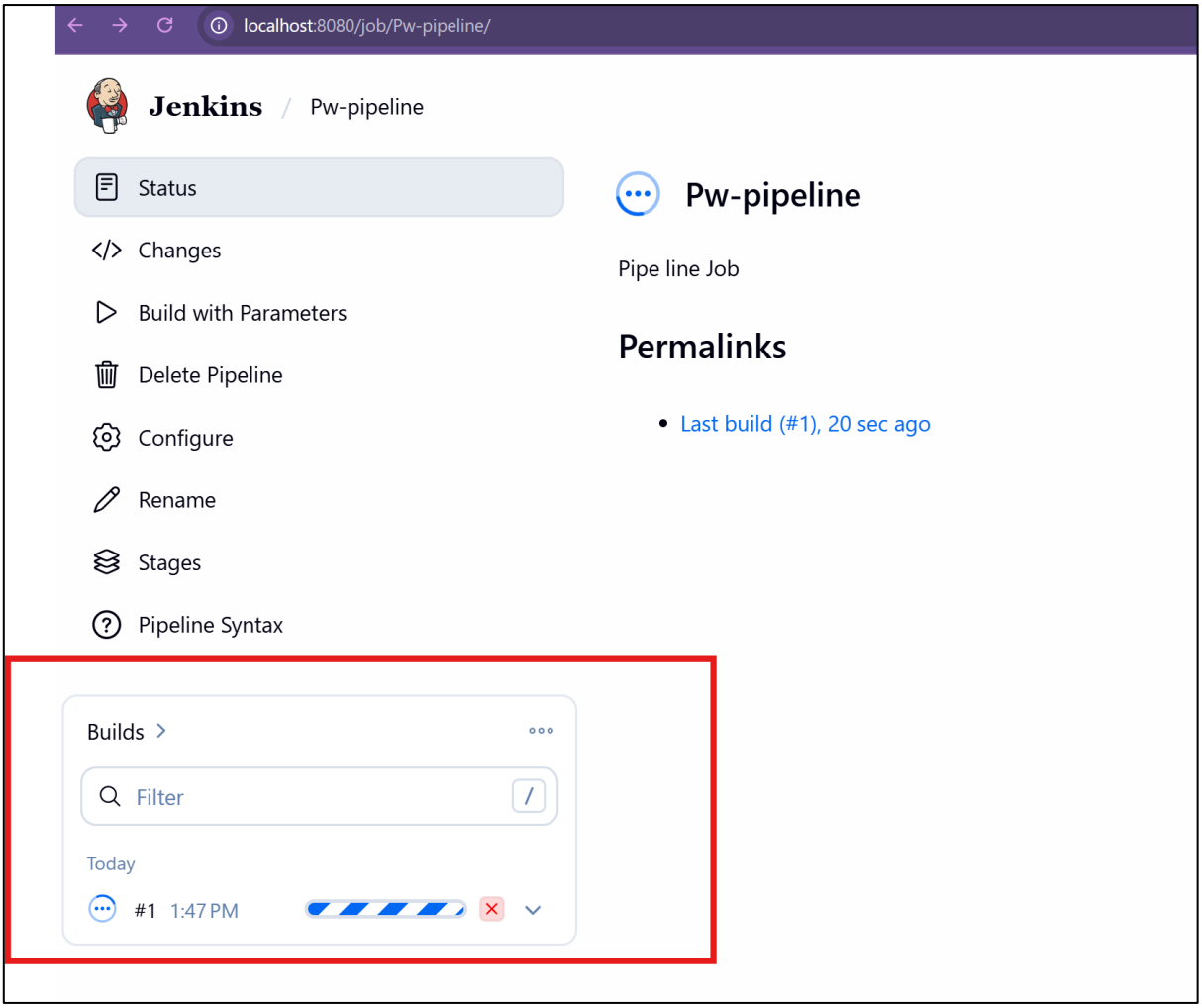


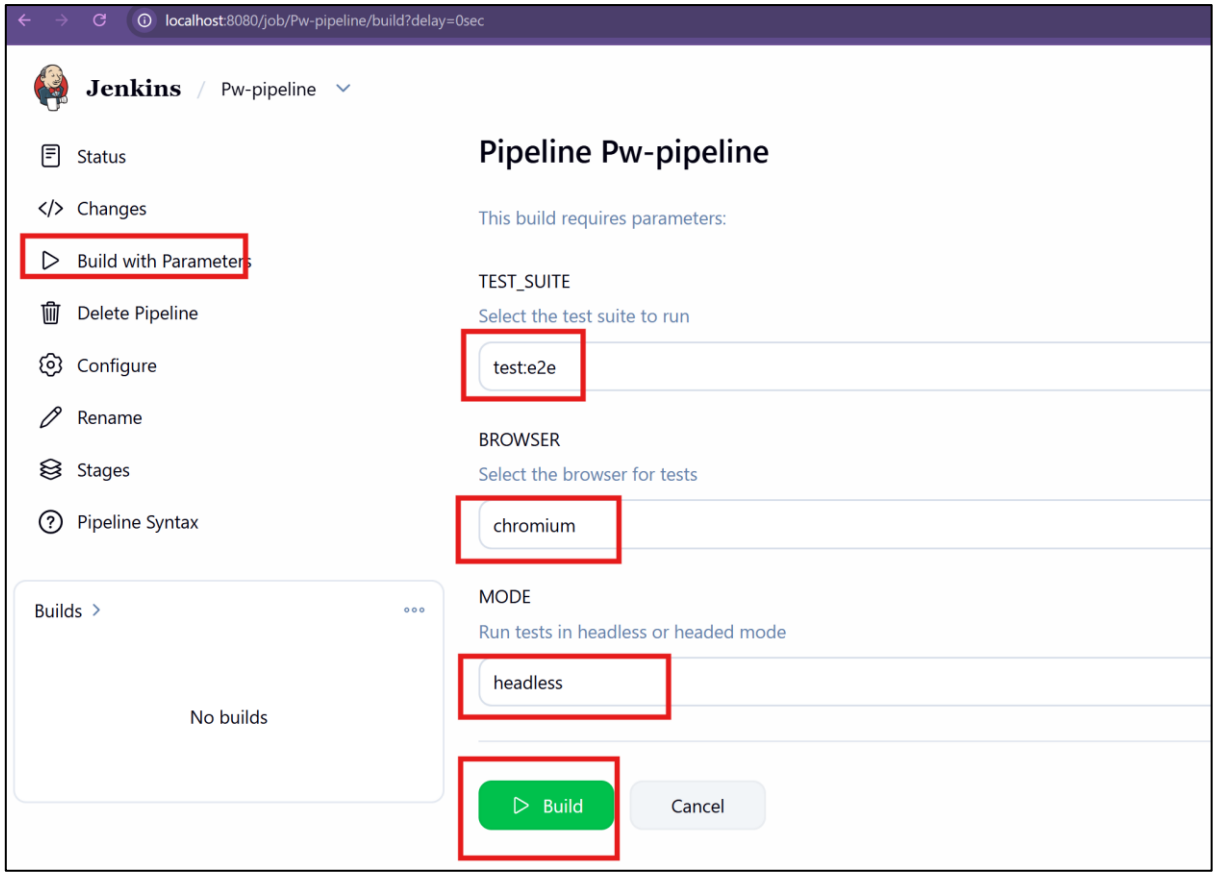


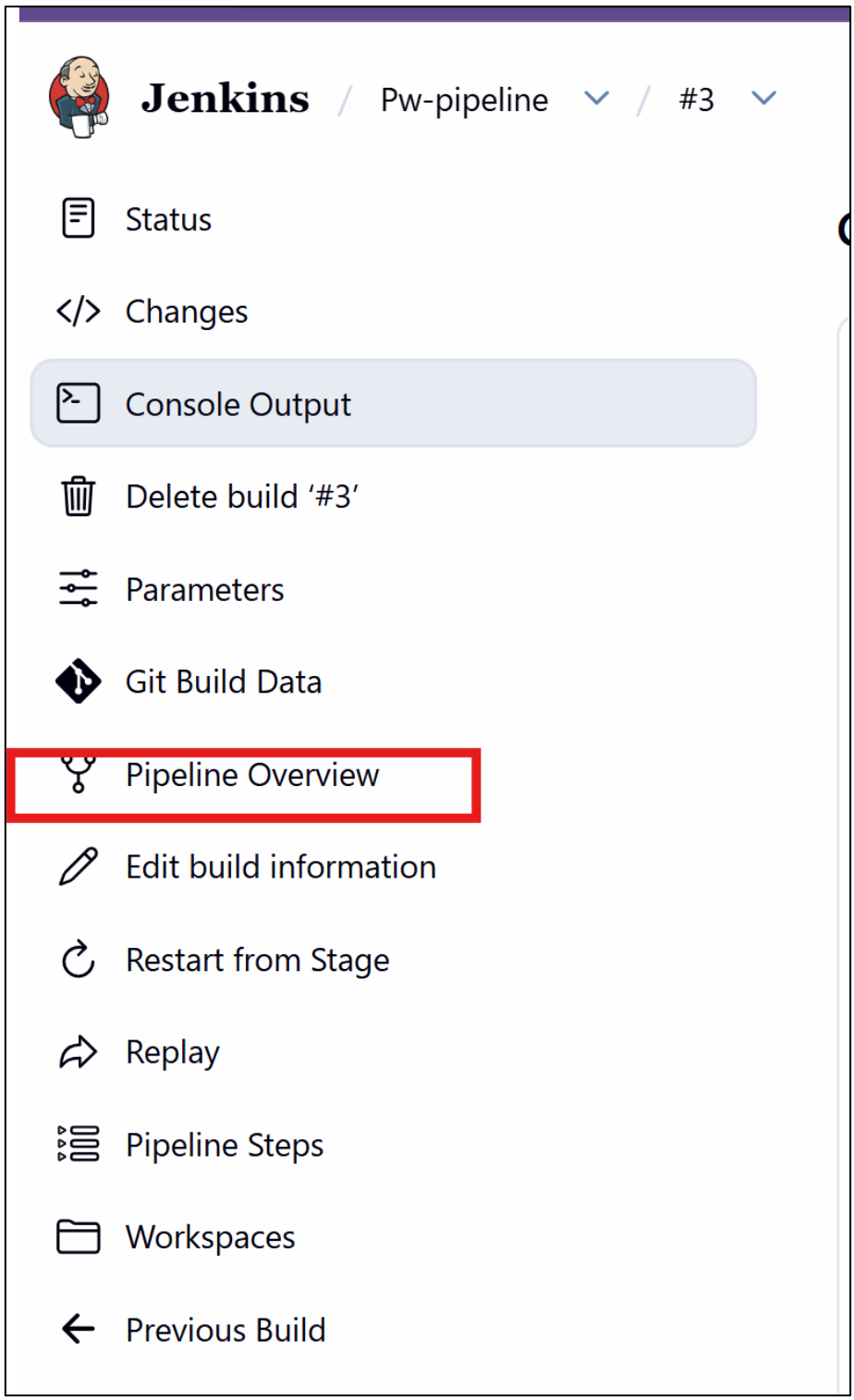


Running a Job




localhost:8080/job/Pw-pipeline/
Jenkins / Pw-pipeline
Status
Changes
Build with Parameters
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Pw-pipeline
Pipe line Job
Permalinks
Last build (#1), 20 sec ago
Builds
Filter
/
Today
#1 1:47 PM

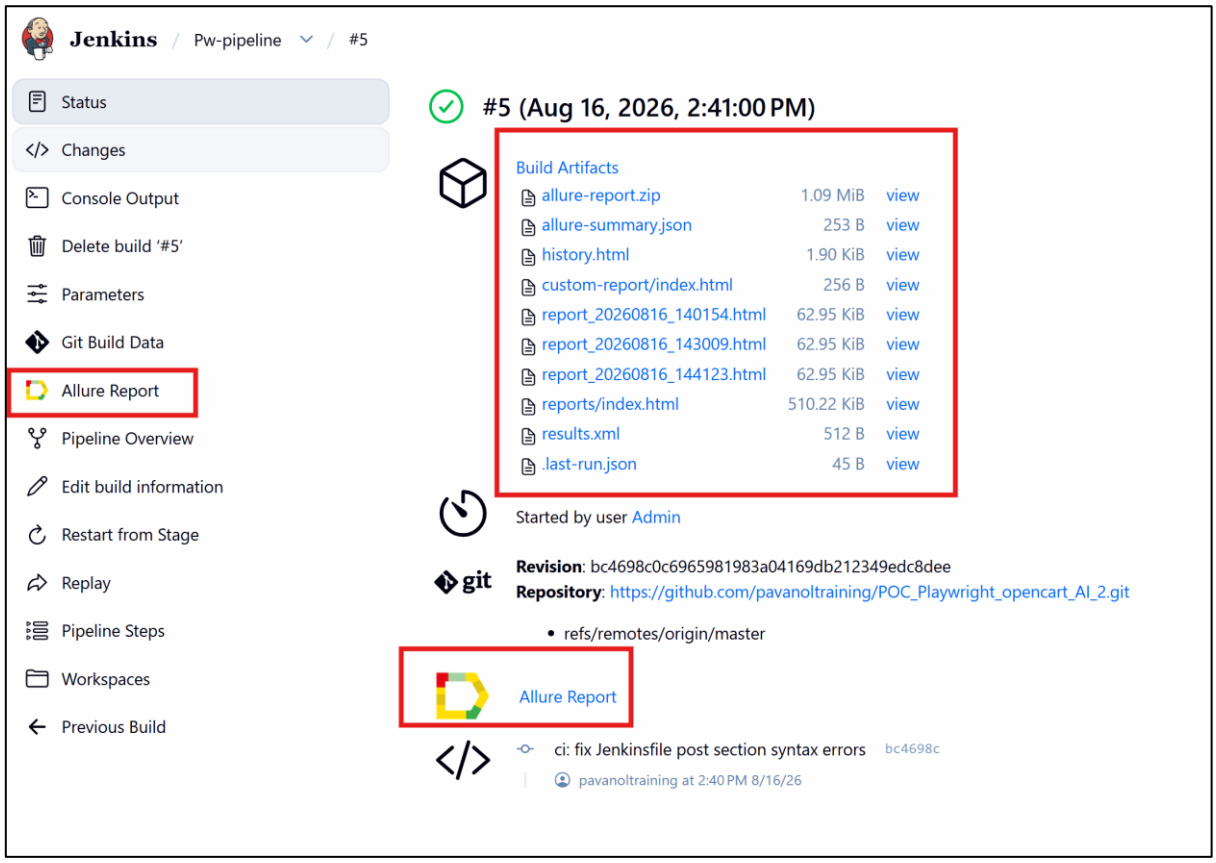

localhost:8080/job/Pw-pipeline/build?delay=0sec
Jenkins
Pw-pipeline
Status
Changes
Build with Parameters
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
No builds
Pipeline Pw-pipeline
This build requires parameters:
TEST_SUITE
Select the test suite to run
test:e2e
BROWSER
Select the browser for tests
chromium
MODE
Run tests in headless or headed mode
headless
Build
Cancel


Jenkins / Pw-pipeline / #3
Status
Changes
Console Output
Delete build '#3'
Parameters
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build

Pipeline Dashboard/Flow:



Results:

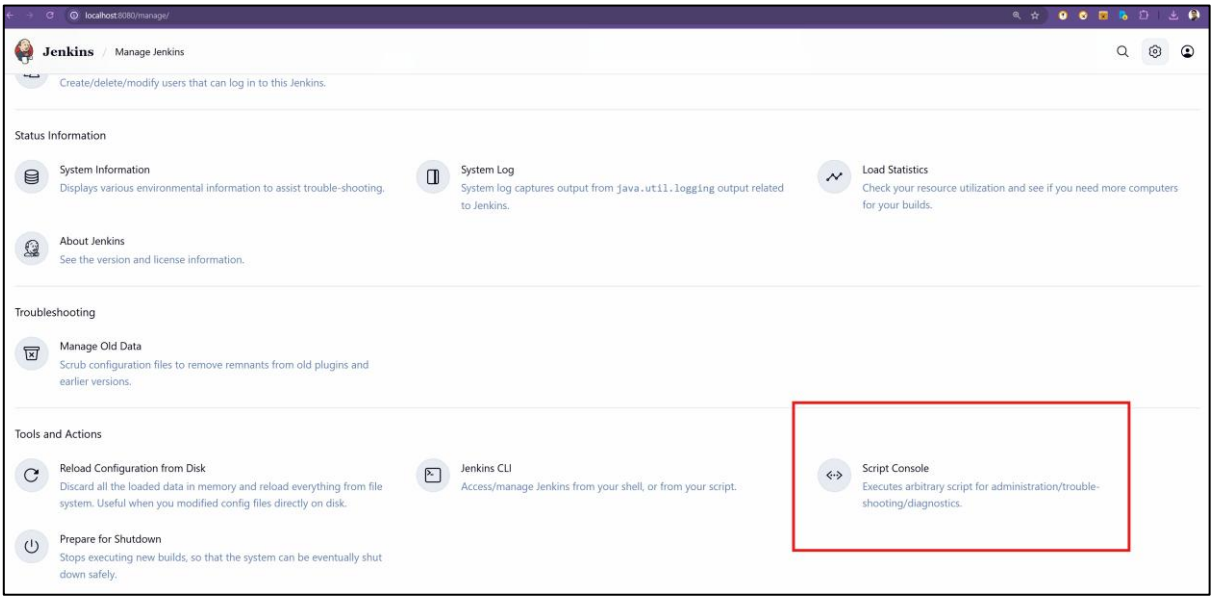


To view HTML Reports, you need to do settings:

** If not visible

Go to:

Manage Jenkins → Script Console



Run below command:

```
System.setProperty("hudson.model.DirectoryBrowserSupport.CSP", "")
```

Reports Available in Jenkins After execution completed:

Report	Source Path	Published Via
Playwright HTML Report	reports/index.html	publishHTML step
Allure Report	allure-results/	allure Jenkins step
Custom Report	custom-report	Archived as artifact
JUnit Test Results	reports/results.xml	junit step
Screenshots / Videos / Traces	custom-report/screenshots/, custom-report/videos/, custom-report/traces/	Archived under custom-report/**

E-mail configuration in Jenkins

Manage Jenkins → System → Extended E-mail Notification

For example, if you use Gmail/Google Workspace:

- SMTP server: smtp.gmail.com
- SMTP port: 465 or 587
- Authentication: use an **App Password**, not your normal Gmail password
- Enable SSL/TLS according to the selected port
- Test the configuration

Then in your **Jenkinsfile**, add a post section.

How to Create the App Password?

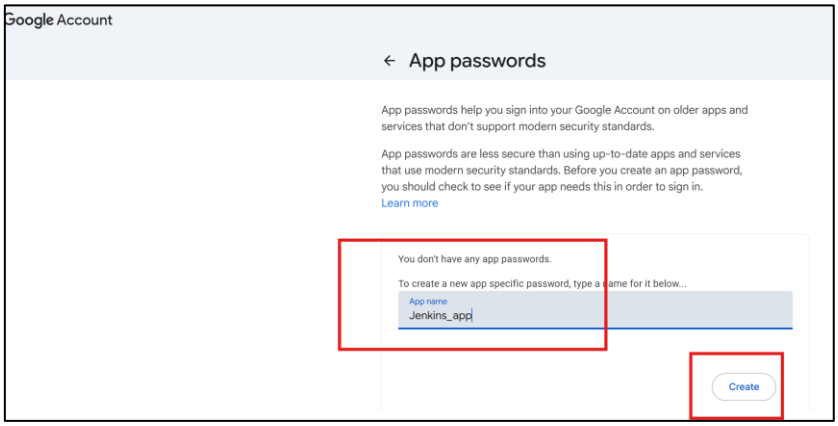
Open: Google App Passwords

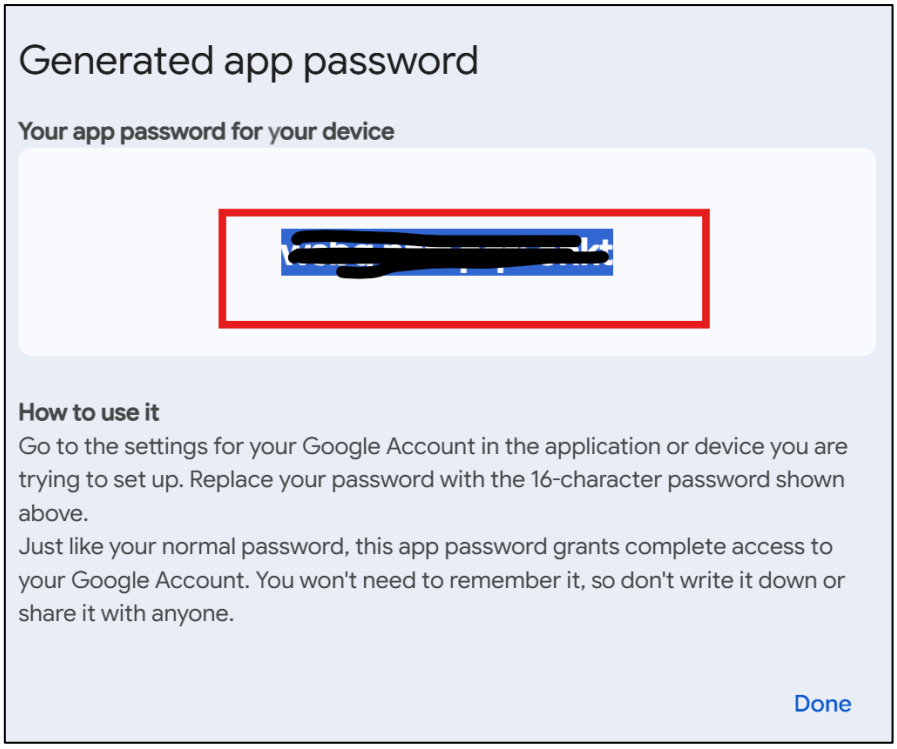
You may be asked to sign in again.

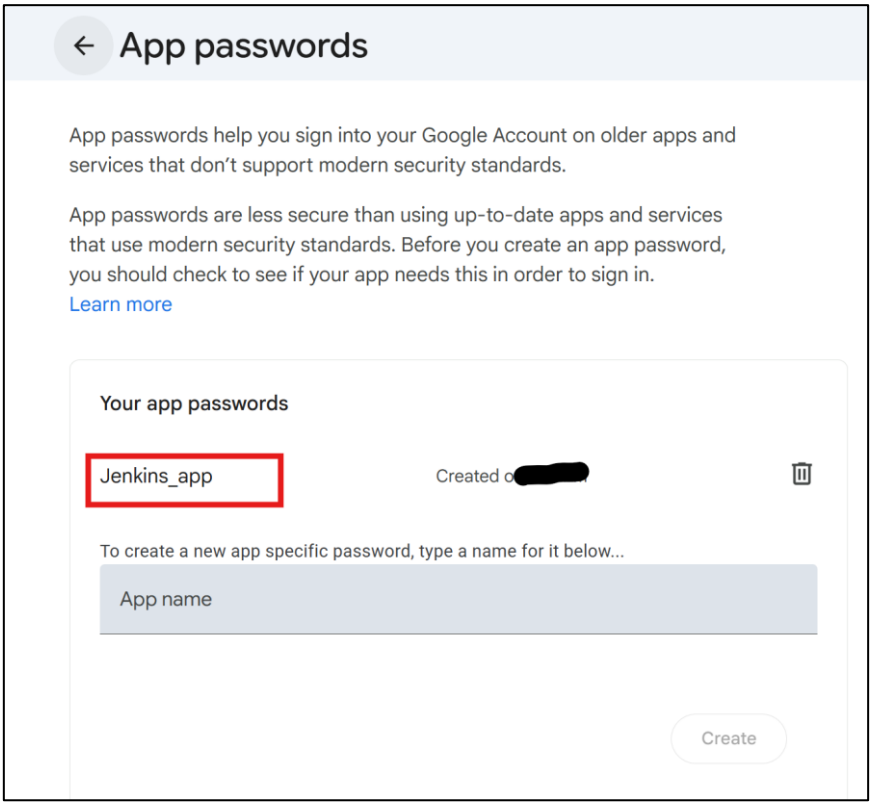
1. Enter an app name, for example:
   Jenkins SMTP
2. Click **Create**.
3. Google will display a **16-character app password**.
4. Copy it immediately.

It will look similar to:

```
abcd efgh ijkl mnop
```




Generated app password
Your app password for your device
How to use it
Go to the settings for your Google Account in the application or device you are trying to set up. Replace your password with the 16-character password shown above.
Just like your normal password, this app password grants complete access to your Google Account. You won't need to remember it, so don't write it down or share it with anyone.
Done


App passwords
App passwords help you sign into your Google Account on older apps and services that don't support modern security standards.
App passwords are less secure than using up-to-date apps and services that use modern security standards. Before you create an app password, you should check to see if your app needs this in order to sign in.
Learn more
Your app passwords
Jenkins_app
To create a new app specific password, type a name for it below...
App name
Create

E-mail Notification configuration in Jenkins:

In Jenkins:

Manage Jenkins → System → E-mail Notification

Use:

SMTP server: smtp.gmail.com

SMTP Port: 587

Username: your-gmail@gmail.com

Password: <16-character App Password>

Use SSL: No

Use TLS: Yes

For port **465**, use SSL instead:

SMTP server: smtp.gmail.com

SMTP Port: 465

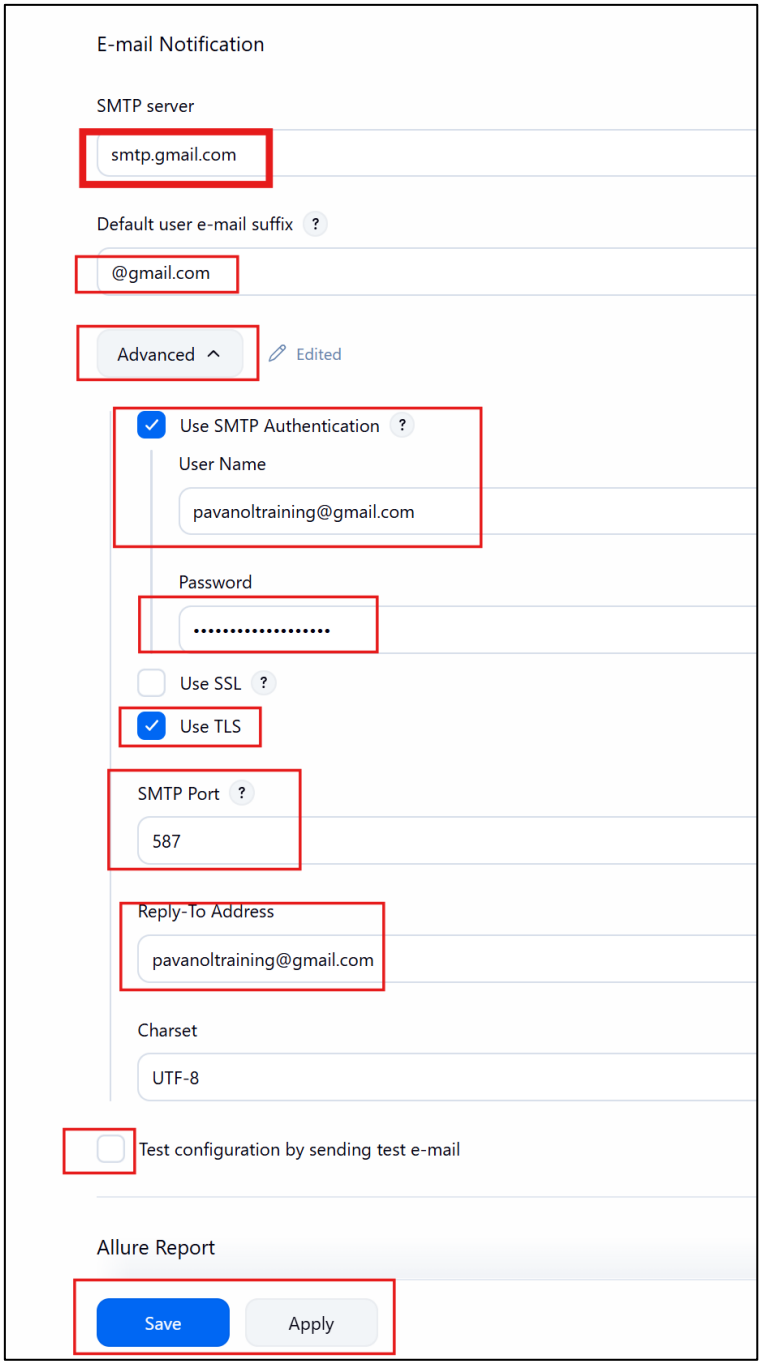
Username: your-gmail@gmail.com

Password: <16-character App Password>

Use SSL: Yes

Use TLS: No

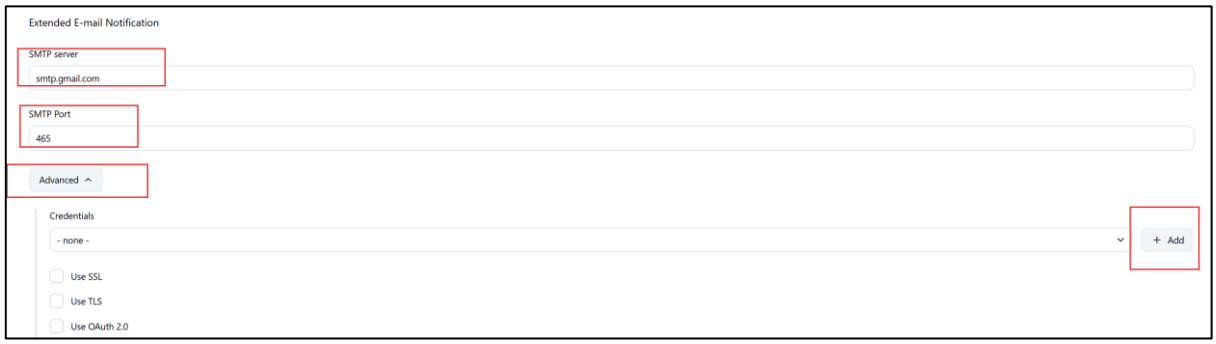
E-mail Notification

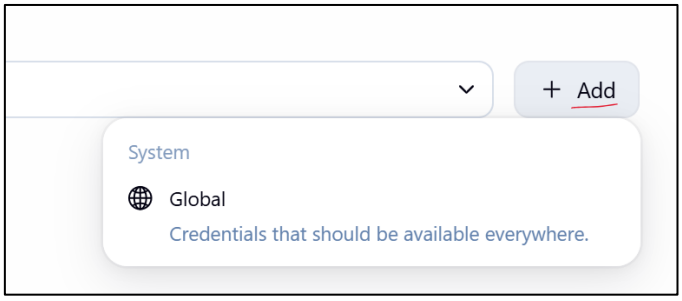


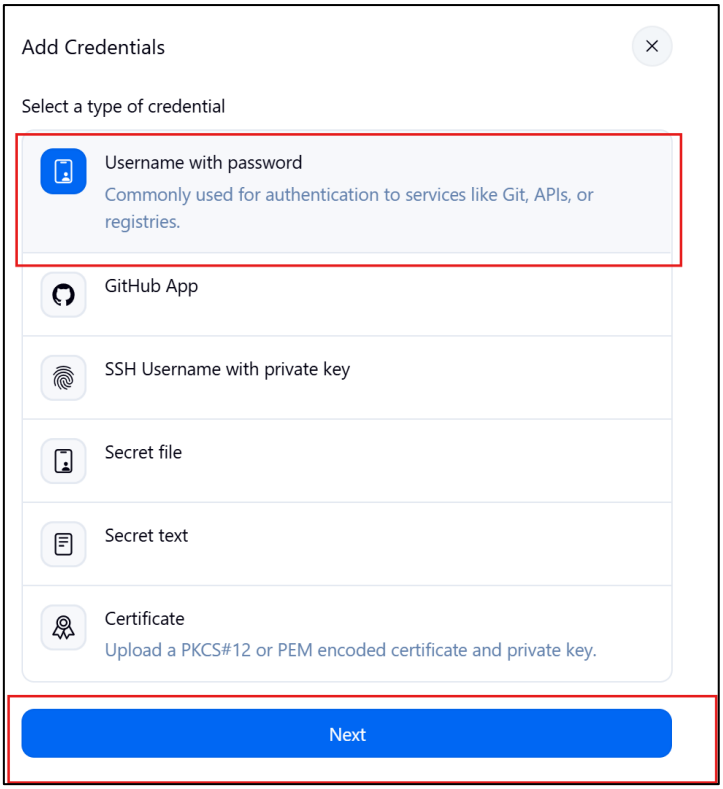


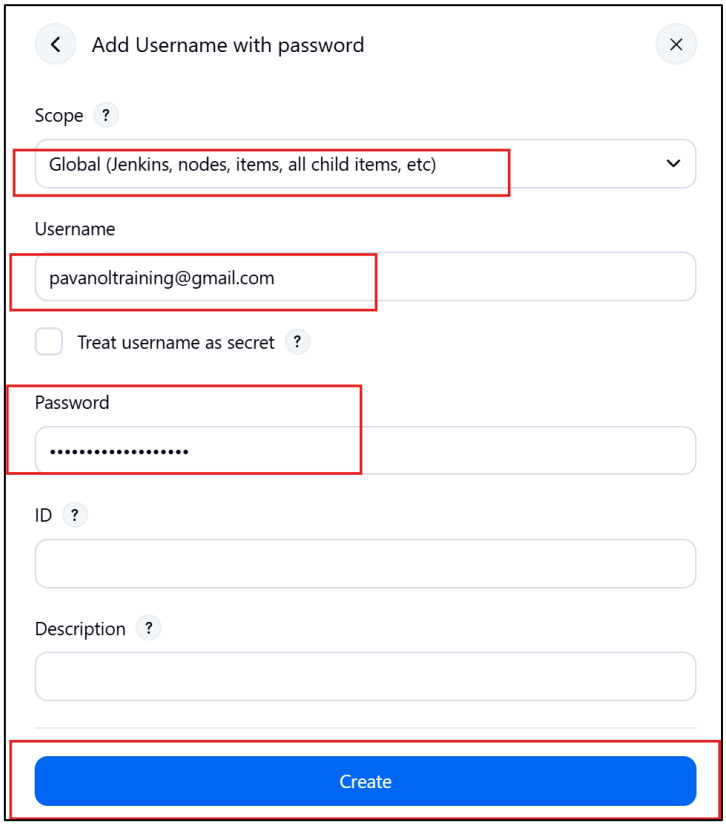


Extended E-mail Notification

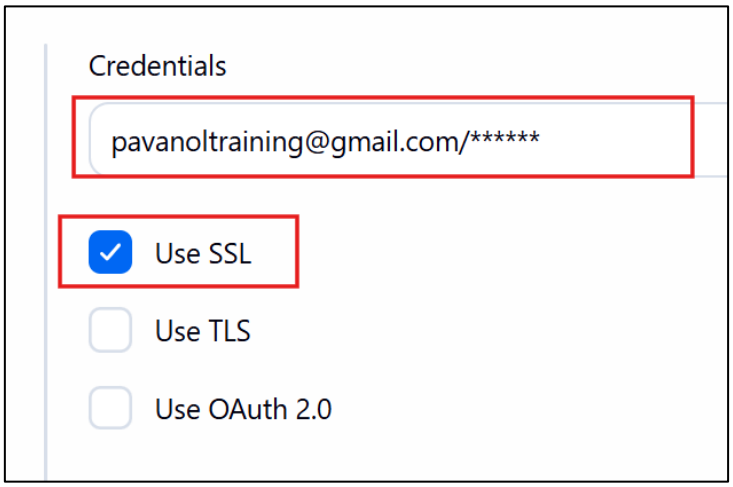





Add Credentials
Select a type of credential
Username with password
Commonly used for authentication to services like Git, APIs, or registries.
GitHub App
SSH Username with private key
Secret file
Secret text
Certificate
Upload a PKCS#12 or PEM encoded certificate and private key.
Next


Add Username with password
Scope
Global (Jenkins, nodes, items, all child items, etc)
Username
pavanoltraining@gmail.com
Treat username as secret
Password
ID
Description
Create





Scheduling Jobs

Generate Cron expression using AI:

https://www.appsmith.com/cron-expression-generator#cron-jobs

http://www.cronmaker.com/?1

